ISM-Summary

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1.Introduction

1.1 Cooling

The primary cooling processes in ISM are two-body radiative processes in which gas loses energy through the emission of photons as a consequence of two-body interactions. There are mainly 4 types of cooling processes.

1. Free-free (bremsstrahlung): Free electron is accelerated by ion. Accelerated charges emit photons, resulting in cooling. Its power is proportional to $T^{1/2}$, which is poorly efficient and only dominant in extremely high temperature environments.
2. Free-bound (recombination): Free electron recombines with ion. Binding energy plus free electron’ s kinetic energy are radiated away. If an electron is captured into anexcited state, subsequent (line) emission may result as electron cascades down to ground level. This requires moderate T_e and slightly high density.
3. Bound-free (collisional ionization): Impact of free electron ionizes a formerly bound electron, taking (kinetic) energy from the free electron.
4. Bound-bound (collisional excitation): Impact of free electron knocks bound electron to excited state. As it decays, it emits a photon. This excludes collisional de-excitation, where no photon is emitted (no net cooling). Forbidden lines are mainly from this process.

To quantify cooling rates, there is volumetric cooling rate ℓ and cooling function Λ. ℓ is the rate per unit volume at which the gas loses energy in erg cm^{-3} s^{-1}, while Λ is a function that is useful when the cooling is achieved by two-body interactions in erg cm^3 s^{-1}. Their relation is $\ell=n_e n_H \Lambda$ when considering the contribution to total cooling rates of different components.

1.2 CLOUDY

CLOUDY is a spectral synthesis code designed to simulate conditions in interstellar matter under a broad range of conditions. Though there are many parameters in CLOUDY, we only need to specify 4 types of inputs.

1. Incident Radiation Fileds: The shape of the spectral energy distribution (SED) and flux intensity or luminosity with distance should be fully specified. CLOUDY computes 3

radiation fields after reading the incident fields. As Fig. 1(a) showing the radiation fields computed in the calculation, the incident radiation field is the external radiation field emitted by the central object that strikes the illuminated face of the cloud. The diffuse radiation field is the radiation field emitted by gas and grains within the nebula. The transmitted radiation field is the net emission emergent from the shielded face of the cloud. It includes both the attenuated incident and the diffuse radiation fields. The reflected radiation field is the emission from the illuminated face of the cloud back into the direction towards (i.e., within 2π sr of) the source of the external field.

2. Chemical Composition: CLOUDY considers 30 elements in detail with all stages of ionization treated. The chemistry can also include many molecules and grains.
3. Density Law: Commands that specify how the hydrogen density is set, and how it changes with radius or depth should be described.
4. Geometry: We can set an open geometry with small covering factor or a closed geometry with the covering factor to a large enough value for light escaping the cloud in the direction towards the central object to always interact gas on the other side. A visualized picture is shown in Fig. 1(b).

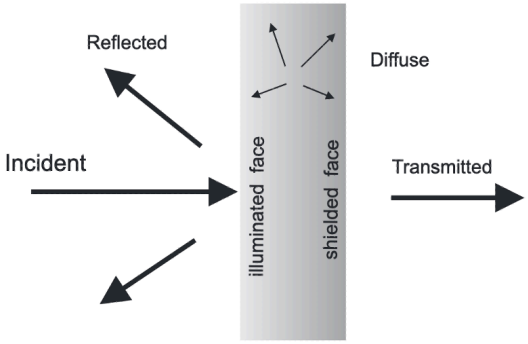


Figure 1(a): Several of the radiation fields that enter in the calculations.

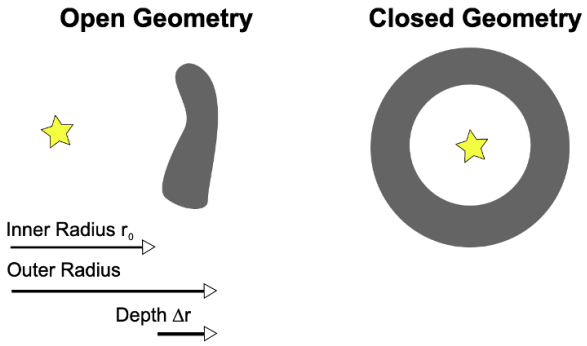


Figure 1(b): This figure shows the two limiting cases that can be assumed. The star is the source of ionizing radiation and the shaded area represents the cloud.

CLOUDY works by dividing a cloud into a set of thin concentric shells referred to as zones. The zones are chosen to have thicknesses that are small enough for the physical conditions across them to be nearly constant. In each zone, CLOUDY will calculate the temperature and all line intensity through ionization balance, thermal balance, excitation balance et. al.

2.AGN Broad Line Region (BLR)

Broad Line Region (BLR) of Active Galactic Nucleus (AGN) is high density, dust-free gas clouds moving at roughly Keplerian velocities at a luminosity dependent distance of 0.01-

1 pc from the central black hole. The typical T ~ 10^4 K, ne > 10^9 cm^{-3} and several times of solar metallicity.

2.1 Depth

We first simulate cooling function with different depths in BLR. We set AGN radiation SED shape and ionization parameter U = 0.01 which is used to set intensity and will be defined later, hydrogen density n(H) = 10^{10} cm^{-3} and 3 times solar metallicity. We use "stop column denisty" command to control the distance in BLR. Fig. 2(a) shows SED in different depths of BLR surface, 1000 km depth and 1AU depth. The AGN radiation is strong in short wave band, and is absorbed as radiation transmits showing mutiple emission and absorption lines. From their cooling and heating functions in Fig. 2(b), we can obtain their equilibrium temperature ~ 10^4 K. Toward deeper distances, the equilibrium temperature decreases, while cooling function increases below 10^6 K. By analysing contributions to total cooling function of 1 AU distance in Fig. 3, we can find H, C, O, Fe, and free-free process dominate cooling function in different temperature section. From CLOUDY outputs, we know collisional lines (bound-bound process) contributes to most of cooling function. Table 1. shows some signifancant lines in coolling, which are all limited lines because of high denisty. Therefore, different temperatures corresponds to diffenert elemet ionization stages and emission lines. Fe lines need more energy while H/C/O are almost fully collisional ionized in ~ 107 K. With all elemnets almost fully ionized in higher temperature, free-free process dominates cooling function. Small depths such as surface environments lead to more ionization and less collisional lines, which can decrease cooling function.

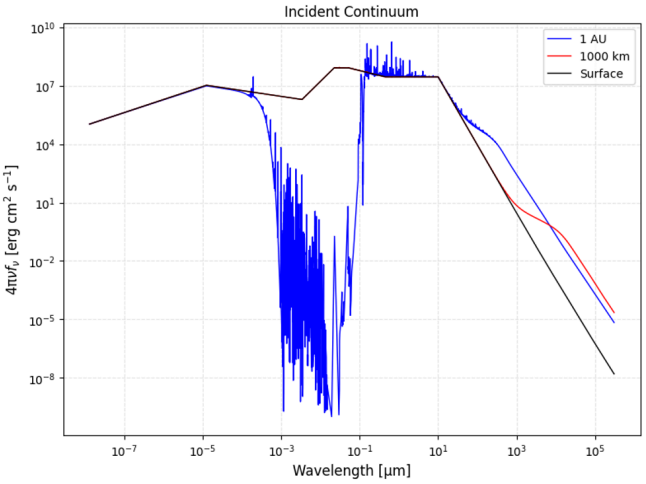


Figure 2(a): AGN SED after transmitting different distances. Black lines are incident radiation fields on BLR surface. Red lines are radiation fields at a depth of 1000 km in BLR, while bule lines are 1 AU depth.

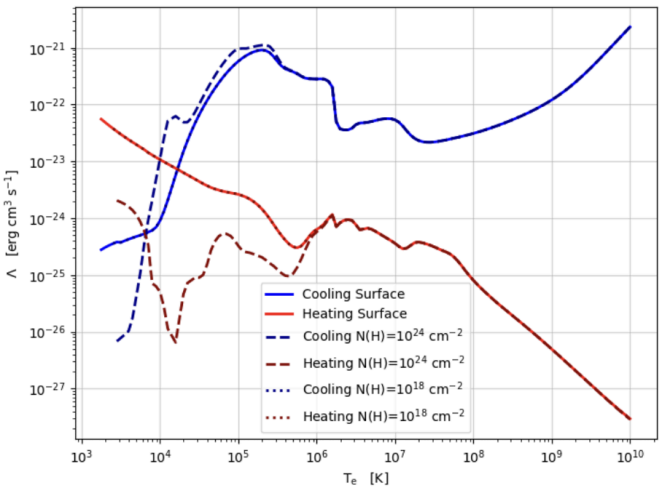


Figure 2(b): Cooling and heating function at different depths. Blue lines are all cooling function and red lines are all heating function. Solid lines are cooling function on BLR surface. Dotted lines are at 1000 km, while dashed lines are at 1 AU.

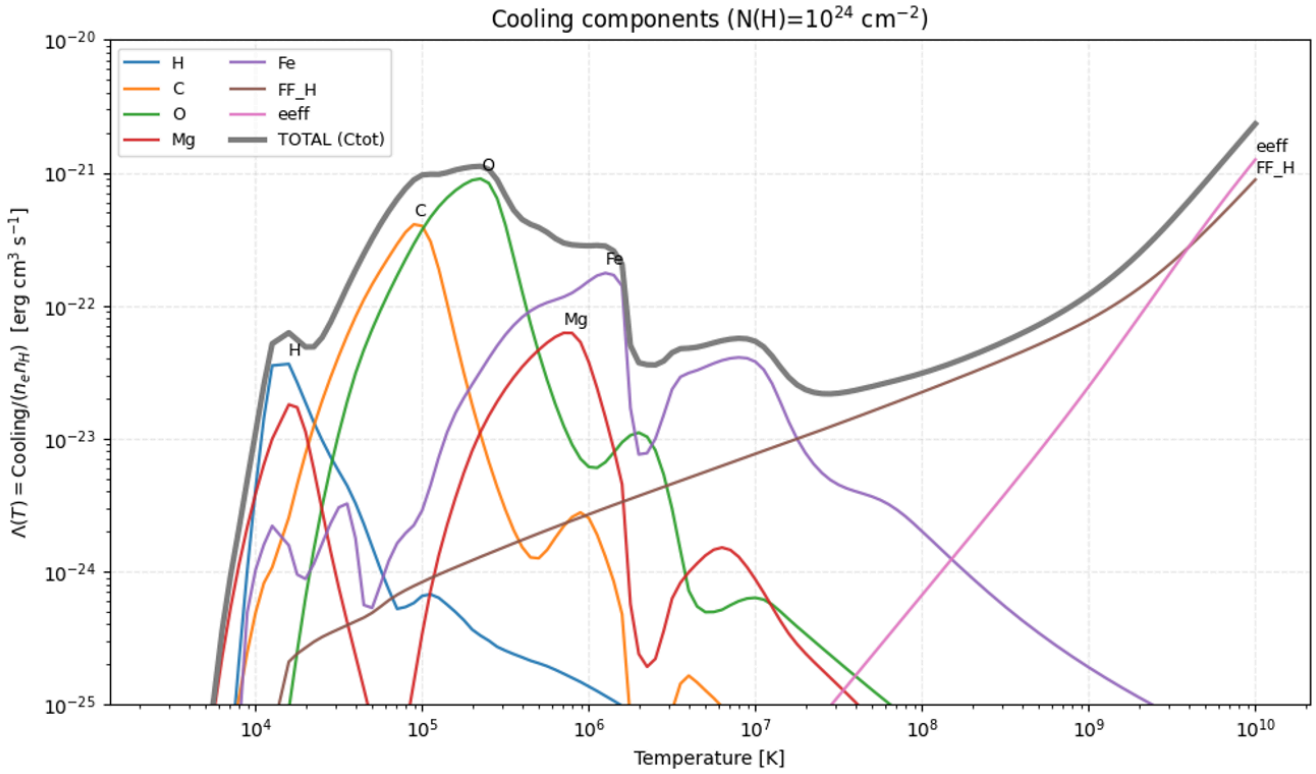


Figure 3: Diiferent components contributing to total cooling function. Gray lines are total cooling function. Blue lines are hydrogen, red lines are magnesium, orange lines are carbon, green lines are oxygen, purple lines are iron, brown lines are electron free-free process and pink line are hydrogen ion free-free process.

Table 1

Significant Lines in Cooling Process

label	Mg 2	Mg 2	H 1	H 1	C 4	C 4	O 6	O 5	Fe 9	Fe11	Fe16	Fe21	Fe23
wave (Å)	2795.53	2802.71	6562.8	1215.67	1548.19	1550.77	1031.91	629.732	171.073	178.48	335.409	128.751	132.906
Energy (eV)	4.44	4.43	1.89	10.22	8.02	8.01	12.04	19.73	72.62	69.61	37.04	96.5	93.48

2.2 Ionization Parameter

Ionization parameter is defined as $U = \frac{Q(H)}{4\pi r^2 n(H) c} = \frac{\Phi(H)}{n(H) c}$, where n(H) [$cm^{-3}$] is the total hydrogen density (ionized, neutral, and molecular) at that point, c is the speed of light, Q(H) [s^{-1}] is the number of hydrogen-ionizing photons emitted by the central object, and Φ(H) [cm^{-2} s^{-1}] is the surface flux of ionizing photons at the illuminated face. We test different ionization parameters on BLR surface and obtained cooling function is shown in Fig. 4. As we can see, larger U leads to lower cooling function because large U corresopnds to more hydrogen-ionizing photons. More H/C/O will exist in fully ionized stage and BLR emits less collisional lines.

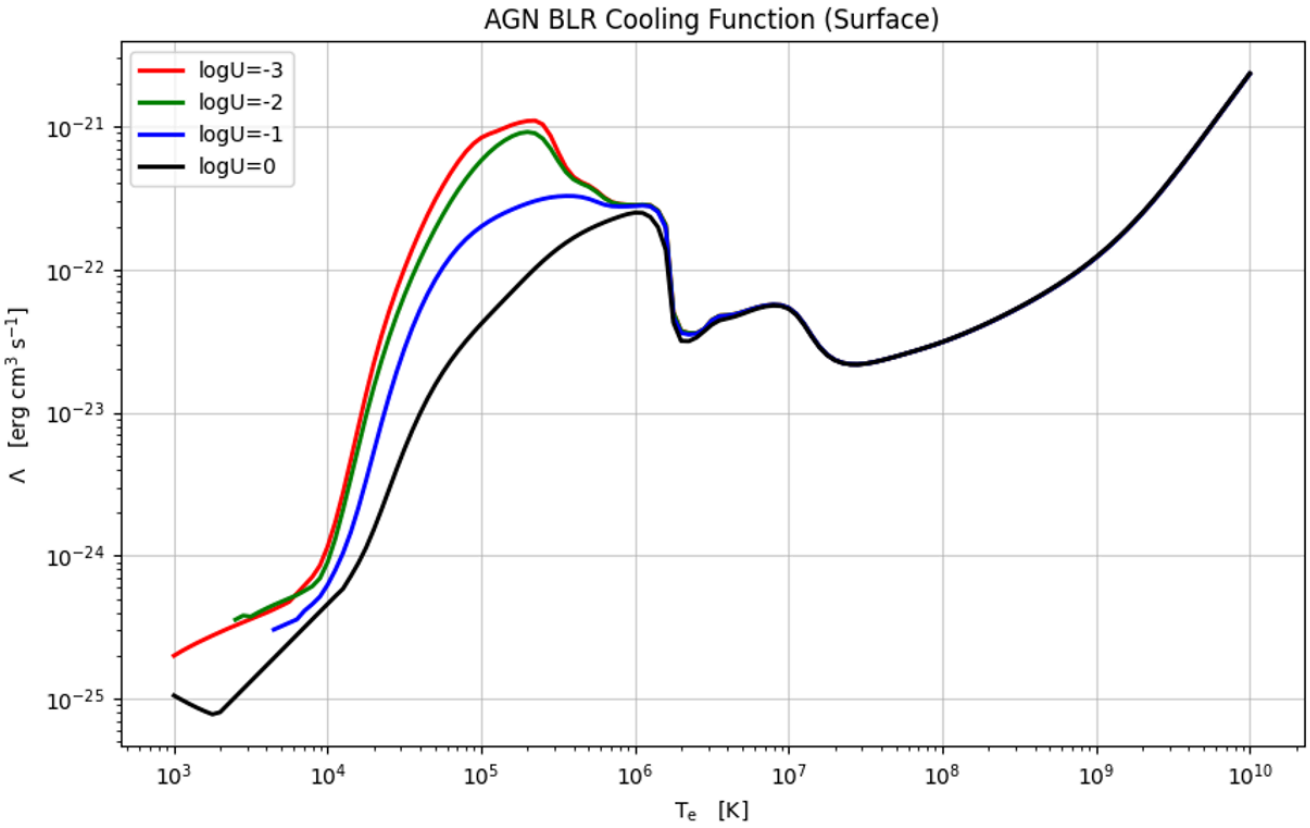


Figure 4: Cooling function of different ionization parameters U on BLR surface. Red lines are U=0.001, green lines are U=0.01, blue lines are U=0.1 and black lines are U=1.

2.3 Thermal Stability

CLOUDY can be used to study thermal stability. Fig. 5 shows T-U/T relation in BLR, which can be intepreted as temperature-pressure phase diagram. In the left region cooling is larger than heating, while in the right heating is larger than cooling. Therefore, in the negative slope region thermal balance is unstable.

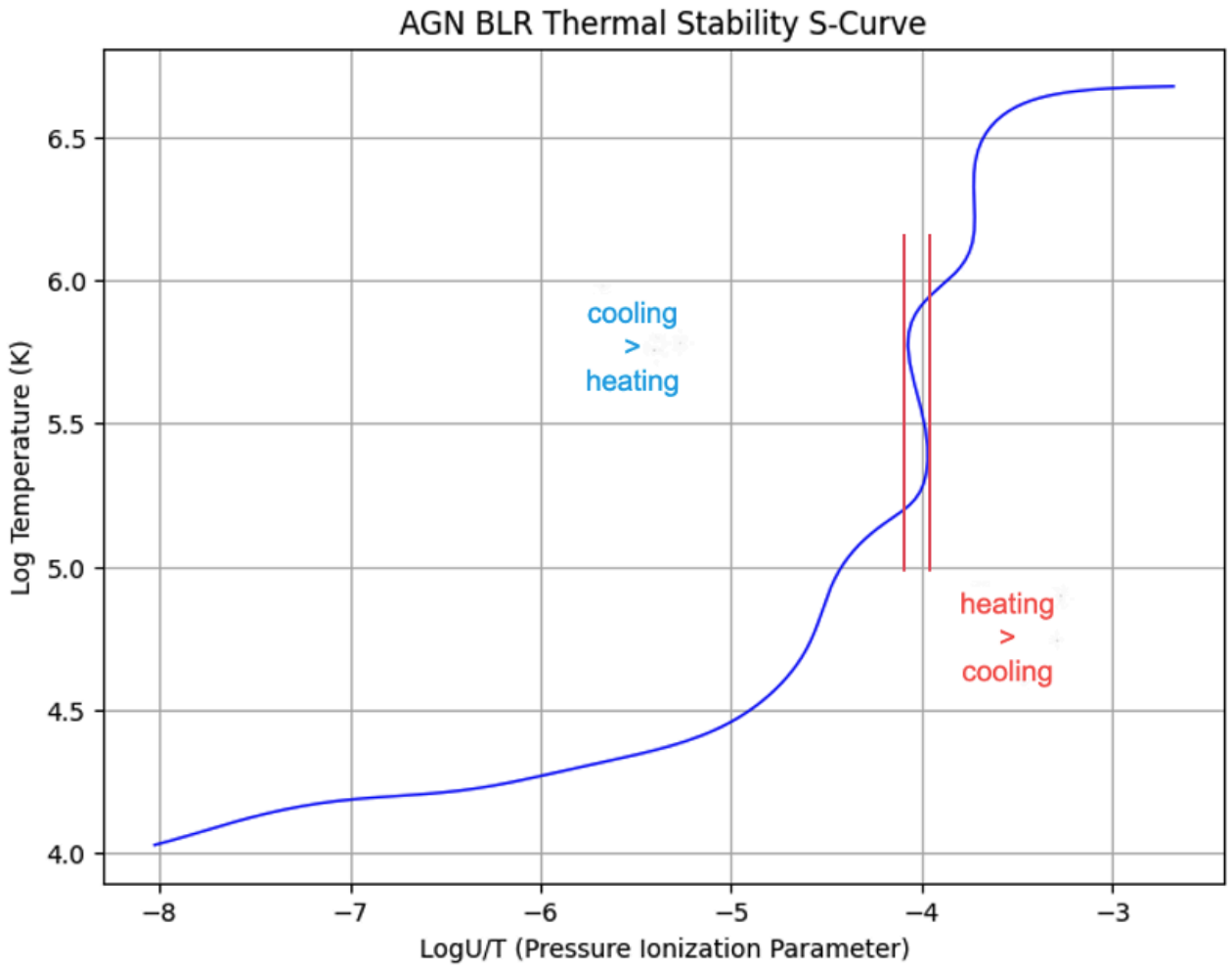


Figure 5: Logarithmic T versus U/T diagram. The blue lines means thermal balance in different conditions. The red lines correspond to unstable region.

2.4 Conclusion

- BLR cooling is mainly based on bound-bound process in relatively low T and free-free process in high T.
- With increasing Te H, Mg, C, O, Fe, ff are the main contributors to cooling.
- Larger depth, lower Φ(H) (i.e. low U) can increase cooling.
- Typical BLR is not stable within logT [K] ~ 5-6

3.HII Region

An H II region is a region of interstellar atomic hydrogen that is ionized. It is typically in a molecular cloud of partially ionized gas in which star formation has recently taken place, with a size ranging from one to hundreds of light years, and density from a few to about a million particles per cubic centimetre. The radiation source is typically an O/B star.

3.1 existence of grains

Fig. 6 shows the cooling function and heating function with and without grains. We can see that grains will significantly improve the cooling rate at high temperatures, this is due to the heat exchange when grains collide with gas. However, in a real physical environment, grains will sublimate at high temperatures. And grains have almost no effect at low temperatures, so we will remove the grains in the following analysis.

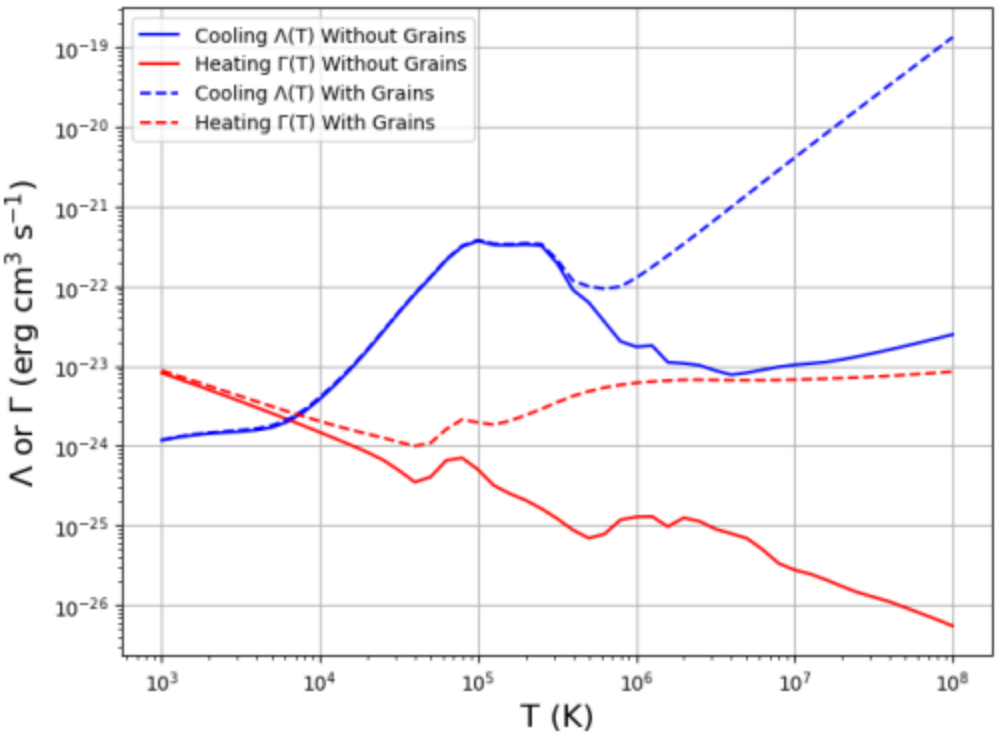


Figure 6: Cooling function (blue curves) and heating function (red curves) with grains (dashed curves) and without grains (solid curves).

3.2 bundance

Fig. 7 shows the total cooling function and the contribution of several main components. We can see that at high temperatures (> 4×10^6 K), free-free process dominates the cooling. At relatively low temperaures, the bound-bound process, i.e. forbidden line of different ionization degrees for different elements dominate the cooling. As the tempereture increases, the dominant forbidden line also changes from a low ionization degree (O II, O III, C III) to a high ionization degree (O VI, Ne VII, Fe IX).

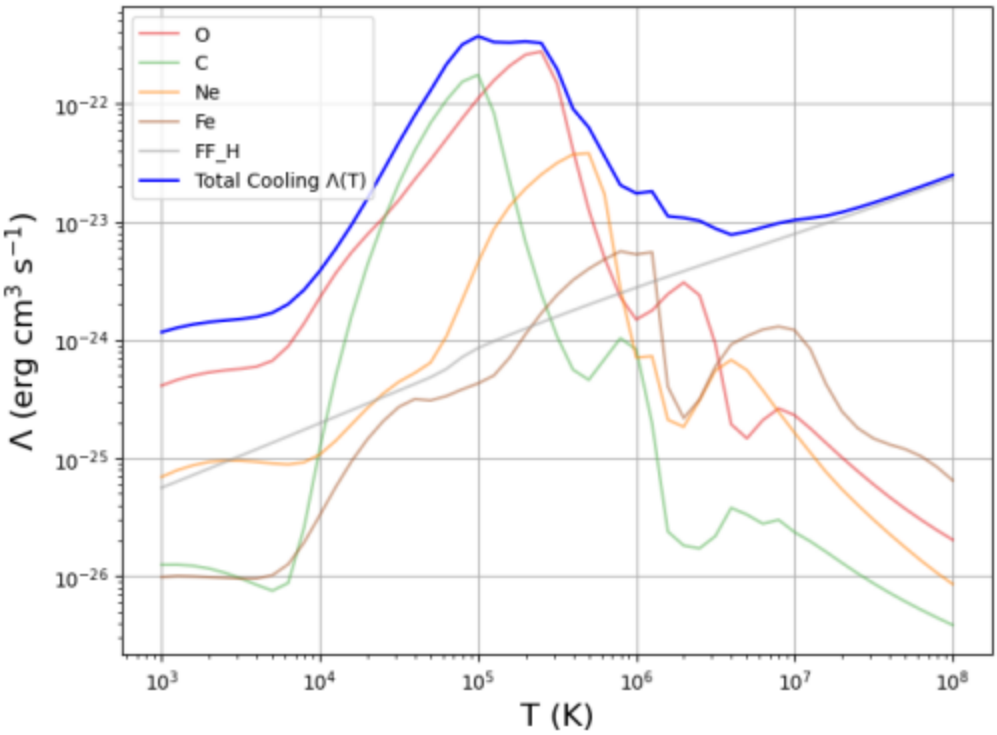


Figure 7: The contribution to cooling function of several main components.

3.3 blackbody temperature

Fig. 8 shows the cooling function of different blackbody temperature of the radiation source. We can see that at high temperatures (> 10^4 K), the blackbody temperature has no effect. This is because the cooling is dominated by collision at high temperatures and by photoionization at low temperatures. High blackbody temperature will cause high ionization degree, and thus improve the cooling rate at low temperatures (< 10^4 K).

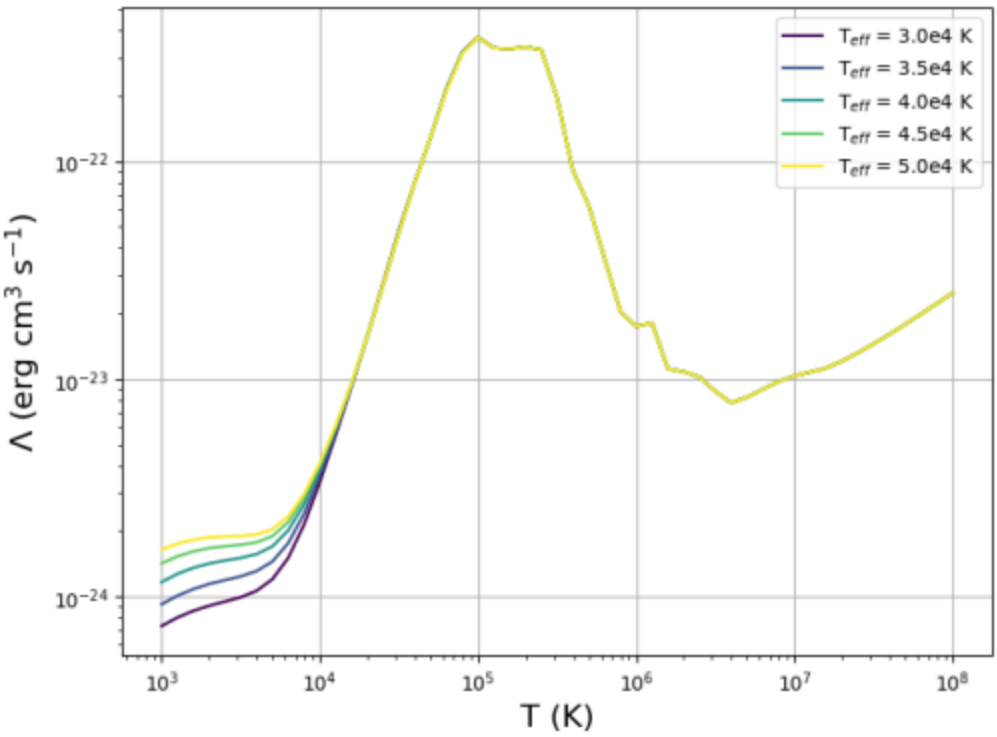


Figure 8: Cooling function of different blackbody temperature of the radiation source.

3.4 ionizing photon flux and hydrogen density

The effect of ionization degree at low temperatures (< 10^4 K) can also be seen at Fig. 9(a), the cooling function of different ionizing photon flux. However, at temperatures between 10^4 K and 4×10^4 K, the cooling rate decreases with increasing ionizing photon flux. This is due to the dominace of hydrogen collisional excitation at this temperature. The stronger ionizing photon flux, the more hydrogen is ionized and the lower the cooling rate. A similar situation also occurs when changing the hydrogen density, as shown in Fig. 9(b). But at low temperatures (< 10^4 K), the low cooling rate with higher hydrogen density is because the high density will suppress the cooling effect of forbidden lines.

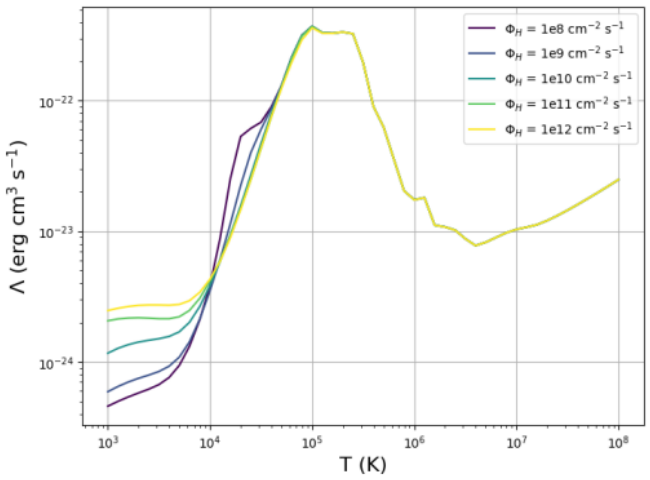


Figure 9(a): Cooling function of different ionizing photon flux.

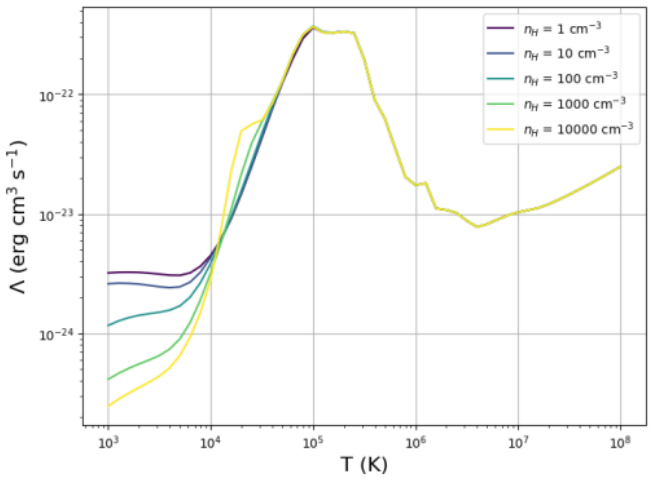


Figure 9(b): Cooling function of different hydrogen density.

3.5 conclusion

There are different dominant cooling process at different temperatures:

- Below 10^4 K, bound-bound process of low ionization forbidden line dominates.
- Between 10^4 K and 4×10^4 K, bound-bound process of hydrogen collisional excitation dominates.
- Between 4×10^4 K and 4×10^6 K, bound-bound process of high ionization forbidden line dominates.
- Above 4×10^6 K, free-free (bremsstrahlung) process dominates.

4.SNR

Supernova remnants (SNRs) are regions of gas and dust produced after a supernova explosion. In this work, we model the SNR as a non-radiative cloud with a spherical geometry, an assumption that is appropriate for most SNR environments.

First, we analyze the cooling contribution of individual elements to identify the dominant cooling agents across different temperature regimes. Second, we investigate the impact of dust grains, metallicity, and cosmic rays on the cooling function. Finally, we perform simplified simulations of different physical regions within the SNR and derive the corresponding cooling functions for each region.

4.1 cooling components

Figure 10(a) and (b) present the dominant cooling elements and their corresponding cooling functions across different temperature regimes. It is evident that different elements dominate the cooling at different temperatures, reflecting the temperature-dependent nature of the cooling processes.

- log (T[K]) below 4, molecule cooling domains.
- log (T[K]) between 4 and 4.5, H cooling domians.
- log (T[K]) between 4.5 and 5, C cooling domians.
- log (T[K]) between 5 and 5.5, O cooling domians.
- log (T[K]) between 5.5 and 6, Ne cooling domians.
- log (T[K]) between 6 and 6.7, S cooling domians.
- log (T[K]) between 6.7 and 7.5, S, Ne, O, H-free free cooling contribute almost equally.
- log (T[K]) above 7.5, H-free free cooling domains.

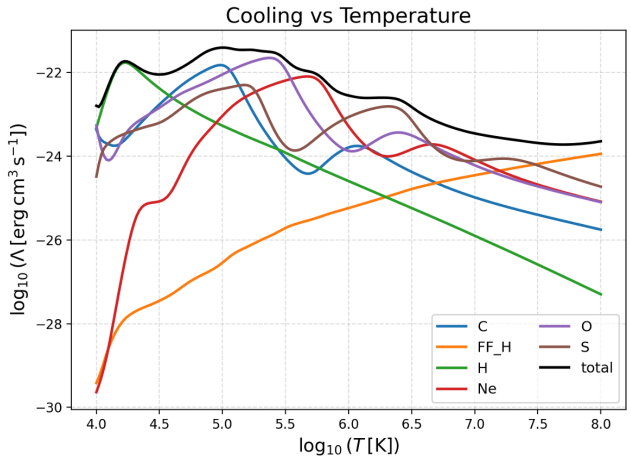


Figure 10(a): Cooling function of several main components

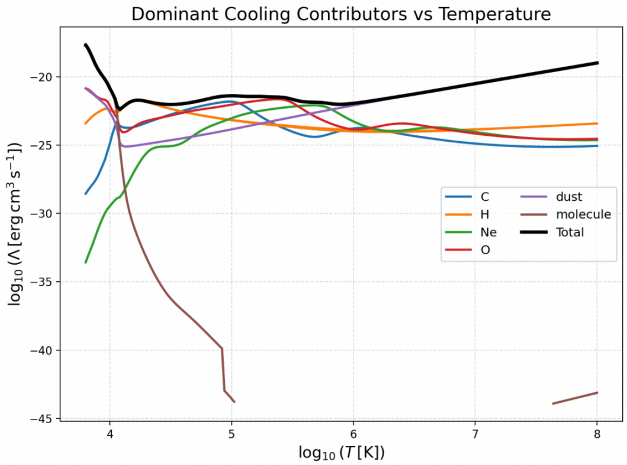


Figure 10(b): Further research about the domian components: cooling function increase in lower temperature due to molecule.

4.2 existence of grains

Figure 11 presents the cooling functions with and without dust grains. The presence of grains enhances the cooling efficiency in both the high-temperature (log [T/K] > 5.5)

and low-temperature (log [T/K] < 4.1) regimes. In the high-temperature regime, the enhanced cooling arises from energy exchange during collisions between gas particles and dust grains, a mechanism analogous to that operating in H II regions (see Section 3.1). In the low-temperature regime, dust grains act as catalytic surfaces for molecule formation, leading to an increased molecular abundance. Since molecular cooling dominates in this temperature range, the increased molecular density results in a significant enhancement of the overall cooling rate.

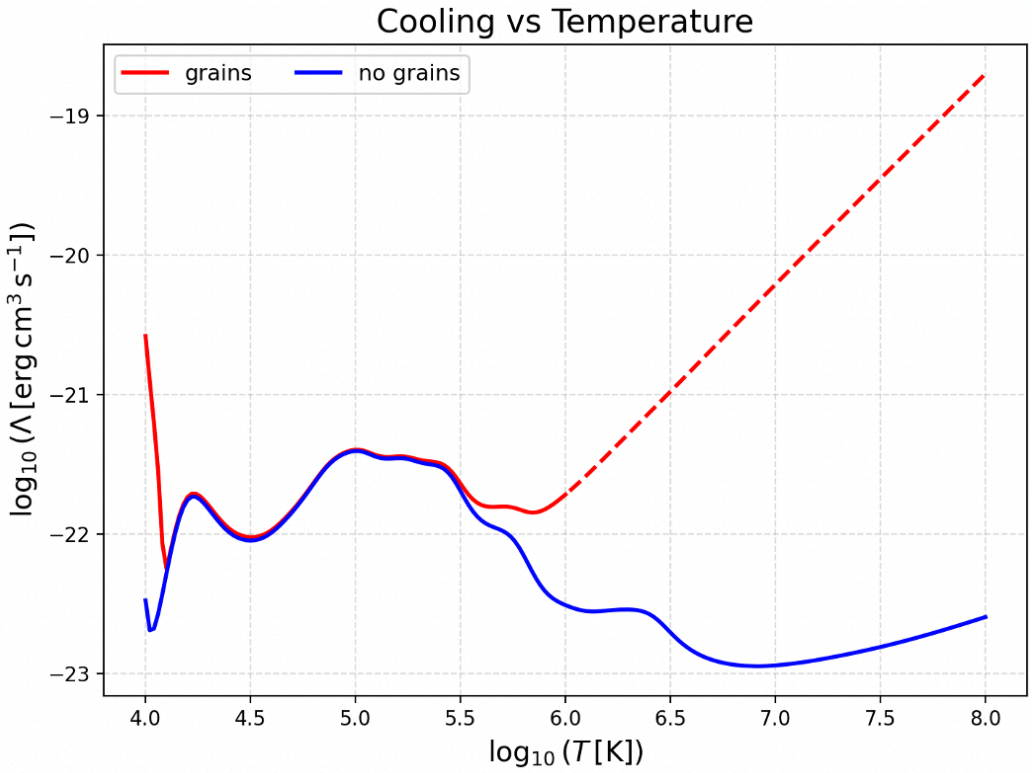


Figure 11: Cooling functions with (red curve) and without (blue curve) dust grains. Since dust grains sublimate at high temperatures in realistic physical environments, the cooling function including grains is shown as a dashed line in the high-temperature regime (log T [K] > 6.0).

4.3 Metallicity

Figure 12 shows the cooling functions for different metallicities. In the temperature range 4.5 < log [T/K] < 6.5, higher metallicity corresponds to a more efficient cooling rate. This behavior arises because metal-line cooling, dominated by elements such as C, O, Ne, and S, is the primary cooling mechanism in this temperature regime.

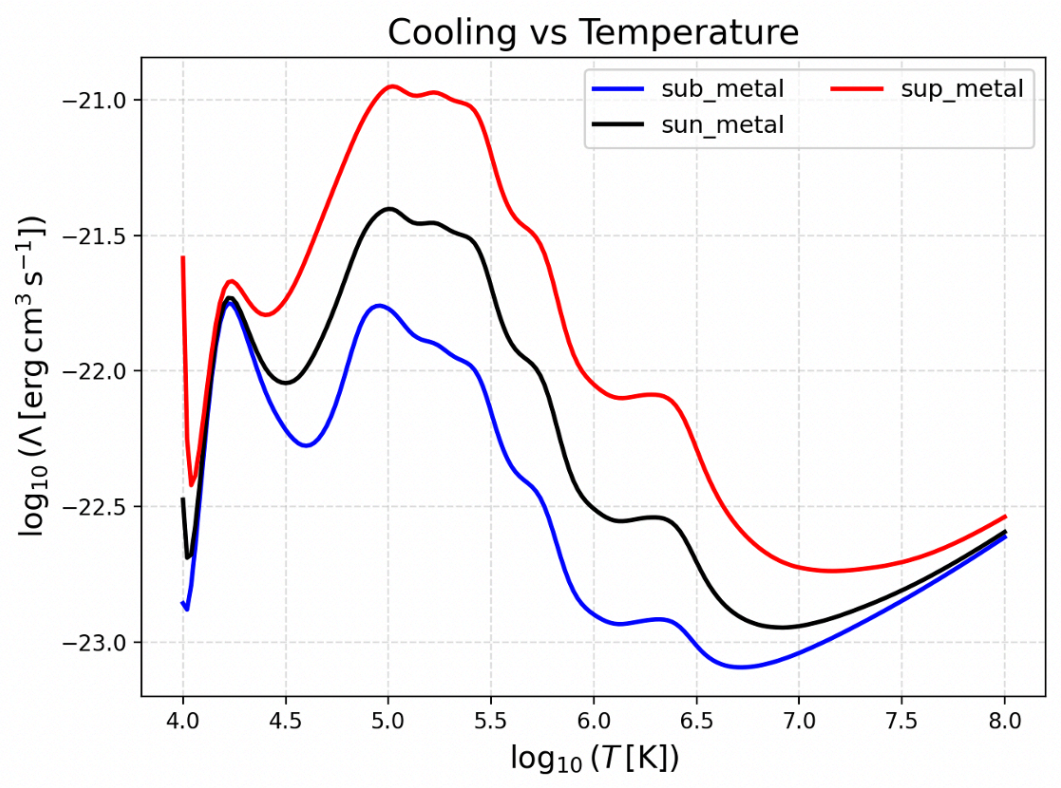


Figure 12: Cooling function with super-metallicity Z~3Zsun(red curve), sun-like metallicity Z~Zsun(black curve) and sub-metallicity Z~0.3Zsun(blue curve).

4.4 Cosmic ray

Figure 13 compares the cooling functions with and without cosmic rays. Across most temperature regimes, cosmic rays have a negligible effect on the cooling function. However, in the low-temperature regime (log T [K] < 4.1), cosmic rays significantly suppress the cooling efficiency. In this temperature range, cooling is dominated by molecular processes, and cosmic rays reduce the cooling rate by destroying molecules through ionization and dissociation, leading to a substantially lower cooling efficiency.

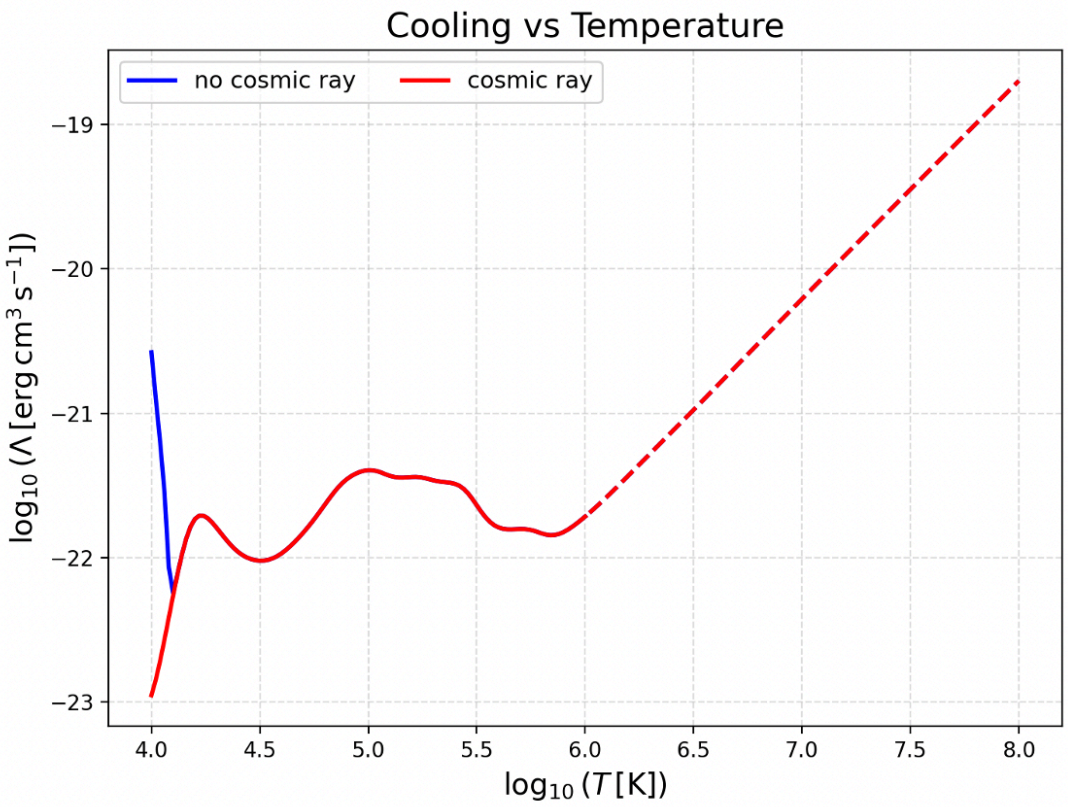


Figure 13: Cooling function with (red curve) and without(blue curve) cosmic ray.

4.5 Simulations of different physical regions

Figure 14 illustrates a simplified physical structure of a supernova remnant (SNR). In this model, two major shocks are present: the forward shock and the reverse shock.

The interstellar medium (ISM) far upstream of the forward shock, referred to as the far-upstream ISM, is essentially unaffected by the SNR evolution and serves as the ambient environment into which the remnant expands. The region immediately behind the forward shock is denoted as the forward-shock shell, where the ISM is strongly compressed by the expanding shock front and cosmic rays are efficiently accelerated. The region between the forward shock and the reverse shock is known as the shocked ejecta. This region is characterized by a relatively lower density due to the free expansion of the ejecta and an enhanced metallicity originating from supernova nucleosynthesis. The innermost region, located inside the reverse shock and referred to as the ejecta core, exhibits an even lower density and a high metal abundance.

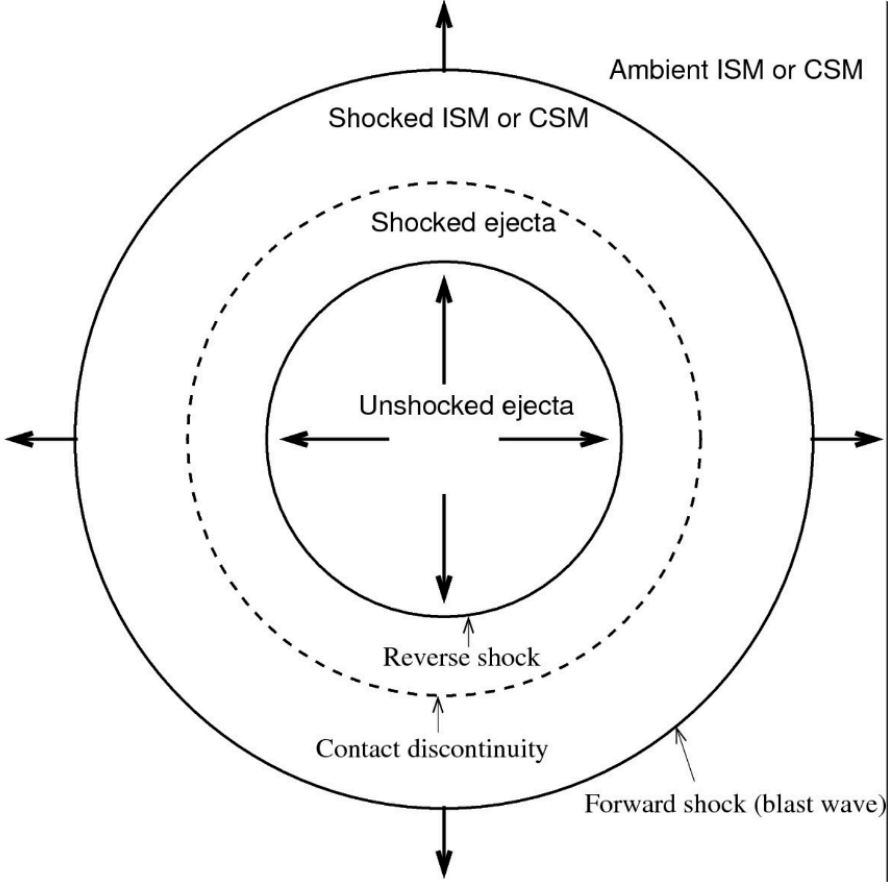


Figure 14: A simplified physical structure of SNR

Figure 15 presents the simulated cooling functions and the relative cooling contributions from several dominant elements in the four SNR regions described above. The simulation parameters are set as follows:

- Far upstream ISM
  - Hydrogen density: 1/cm^3
  - Cosmic rays: background
  - abundances: ISM
- Forward shock shell
  - Hydrogen density: 10/cm^3
  - Cosmic rays: background×10
  - abundances: ISM
- Shocked ejecta
  - Hydrogen density: 0.1/cm^3
  - Cosmic rays: background
  - abundances: ISM, but metallicity Z~10Zsun
- Ejecta core
  - Hydrogen density: 0.01/cm^3
  - Cosmic rays: background
  - abundances: ISM, but metallicity Z~10Zsun

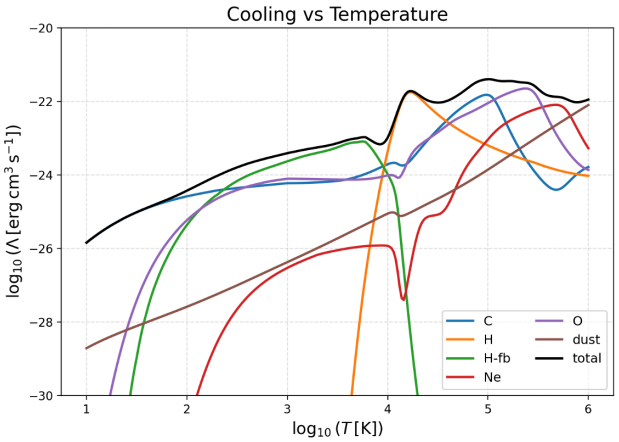


Figure 15(a): Cooling function of far upstream ISM

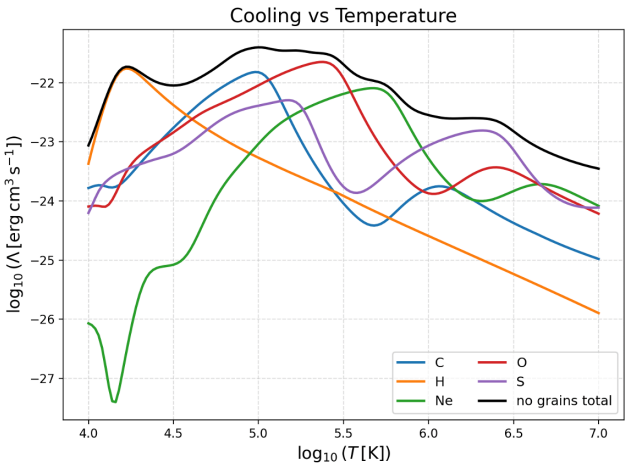


Figure 15(b): Cooling function of forward shock shell

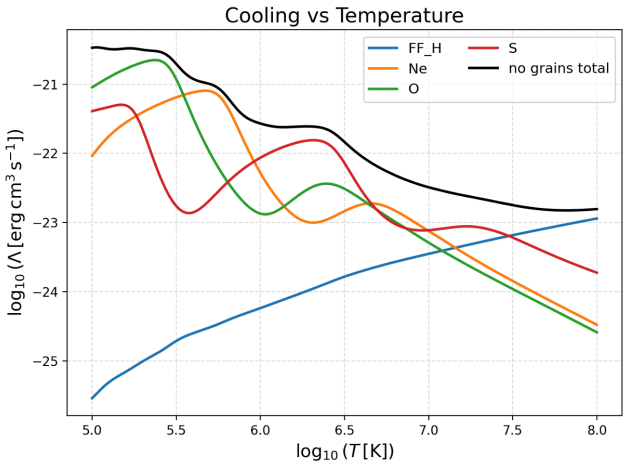


Figure 15(c): Cooling function of shock ejecta

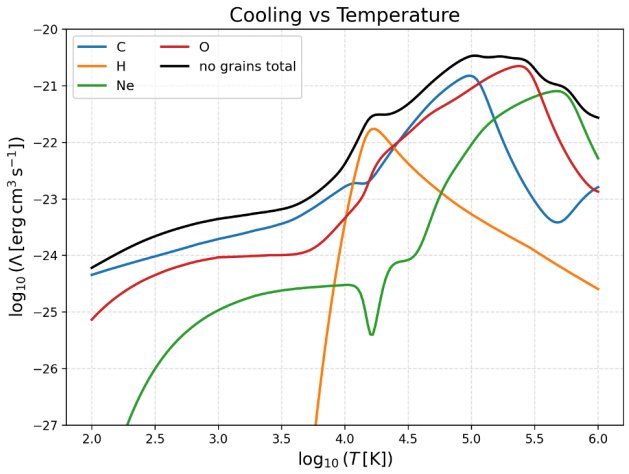


Figure 15(d): Cooling function of ejecta core

4.6 Conclusion

- Across the temperature range from low (log T [K] < 4) to high (log T [K] > 8), the dominant cooling agents in SNRs transition sequentially from molecules to atomic hydrogen, followed by metals including C, O, Ne, and S, and finally to hydrogen free–free emission at the highest temperatures.
- Dust grains enhance the cooling efficiency in both the low-temperature regime (log T [K] < 4.1) and the high-temperature regime (log T [K] > 5.5), while their contribution is negligible at intermediate temperatures.
- Higher metallicity leads to a significantly increased cooling rate in the temperature range 4.5 < log (T/K) < 6.5, where metal-line emission dominates the cooling process.
- Cosmic rays suppress the cooling rate in the low-temperature regime (log T [K] < 4.1), where molecular cooling is dominant.

5.Task Allocation

The usage of CLOUDY is studied by all of us, and parameters of CLOUDY are mainly summarized by Shuaiyi Li. We then probe cooling function in three different ISM conditions. Shuaiyi Li studies AGN BLR, Jingshuo Yang researches HII region and Hongyu Li explores SNR.